1. Executive Summary

This report presents strategic insights derived from an end-to-end analytics workflow conducted over three weeks.
The objective was to understand learner engagement behavior, identify drivers of application trends and student churn, and propose data-driven actions to improve retention.

The project integrates data preparation and feature engineering, exploratory data analysis, and predictive churn modeling to support operational and strategic decision-making.

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2. Data Foundation and Analytical Scope

During Week 1, the raw learner–opportunity dataset was cleaned, validated and transformed into an analysis-ready structure.

Key engineered features included:

- learner age
- opportunity duration
- signup month and year
- days between signup and application

These features enabled temporal trend analysis, demographic segmentation and engagement modeling.

```
 ++ Feature Engineering ++
Calculating Age ==>
     Learner SignUp DateTime  ...   Age
0        2023-06-14 12:30:35  ...  21.0
1        2023-01-05 05:29:16  ...  22.0
2        2023-09-04 20:35:08  ...  21.0
3        2023-08-29 05:20:03  ...  24.0
4        2023-06-01 15:26:36  ...  23.0
```

```
Opportunity Duration ==>
     Learner SignUp DateTime  ... Opportunity_Duration_Days
0        2023-06-14 12:30:35  ...                     841.0
1        2023-01-05 05:29:16  ...                     841.0
2        2023-09-04 20:35:08  ...                     841.0
3        2023-08-29 05:20:03  ...                     841.0
4        2023-06-01 15:26:36  ...                     841.0
```

```
Signup Month & Year ==>
     Learner SignUp DateTime  ... Signup_Year
0        2023-06-14 12:30:35  ...      2023.0
1        2023-01-05 05:29:16  ...      2023.0
2        2023-09-04 20:35:08  ...      2023.0
3        2023-08-29 05:20:03  ...      2023.0
4        2023-06-01 15:26:36  ...      2023.0
```

```
Application Delay Feature ==>
     Learner SignUp DateTime  ... Days_To_Apply
0        2023-06-14 12:30:35  ...           0.0
1        2023-01-05 05:29:16  ...           0.0
2        2023-09-04 20:35:08  ...           NaN
3        2023-08-29 05:20:03  ...          12.0
4        2023-06-01 15:26:36  ...           0.0
```

3. Engagement and Application Behavior Insights

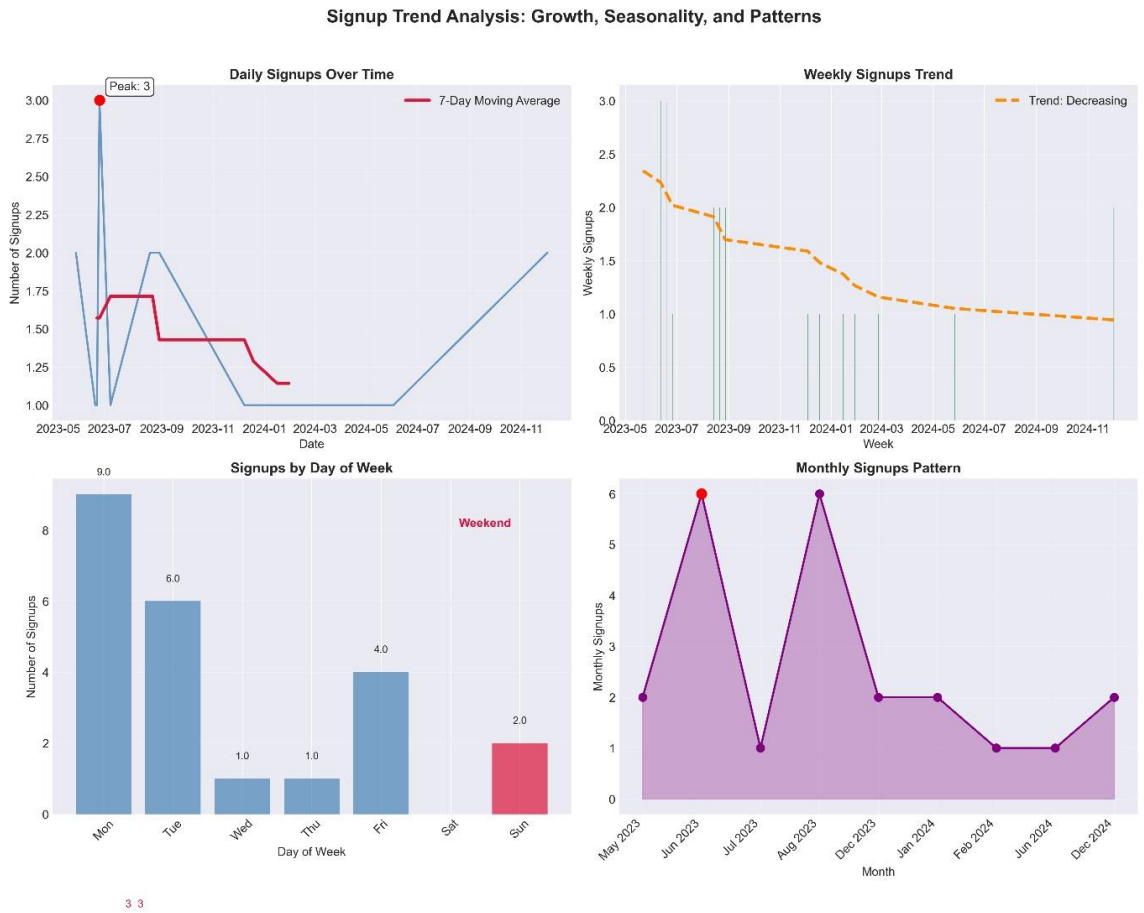
Exploratory analysis reveals several important behavioral patterns.

The learner population is dominated by young adults, with engagement behavior showing strong right-skewness.
A small group of learners exhibits very high long-term engagement, while the majority remain moderately active.

Gender participation is imbalanced, indicating potential outreach or accessibility gaps.

However, the analytical dataset was significantly reduced after cleaning, highlighting important data quality risks that must be addressed before production deployment.

➞ Age distribution
➞ Gender distribution
➞ Engagement days distribution
➞ Engagement months distribution
➞ Before vs after cleaning record count


Signup Trend Analysis: Growth, Seasonality, and Patterns
Daily Signups Over Time
Peak: 3
7-Day Moving Average
Number of Signups
Date
Weekly Signups Trend
Trend: Decreasing
Weekly Signups
Week
Signups by Day of Week
Weekend
9.0
6.0
1.0
1.0
4.0
2.0
Number of Signups
Mon
Tue
Wed
Thu
Fri
Sat
Sun
Day of Week
Monthly Signups Pattern
Monthly Signups
May 2023
Jun 2023
Jul 2023
Aug 2023
Dec 2023
Jan 2024
Feb 2024
Jun 2024
Dec 2024
Month

Completion Rate Trend vs Signup Volume
Completion Rate (%)
Trend: Declining (0.00%/week)
Weekly Signups
Highest: 0.0%
Lowest: 0.0%
Completion Rate (%)
Number of Signups
Week
Correlation: nan
(Negative = more signups → lower completion)

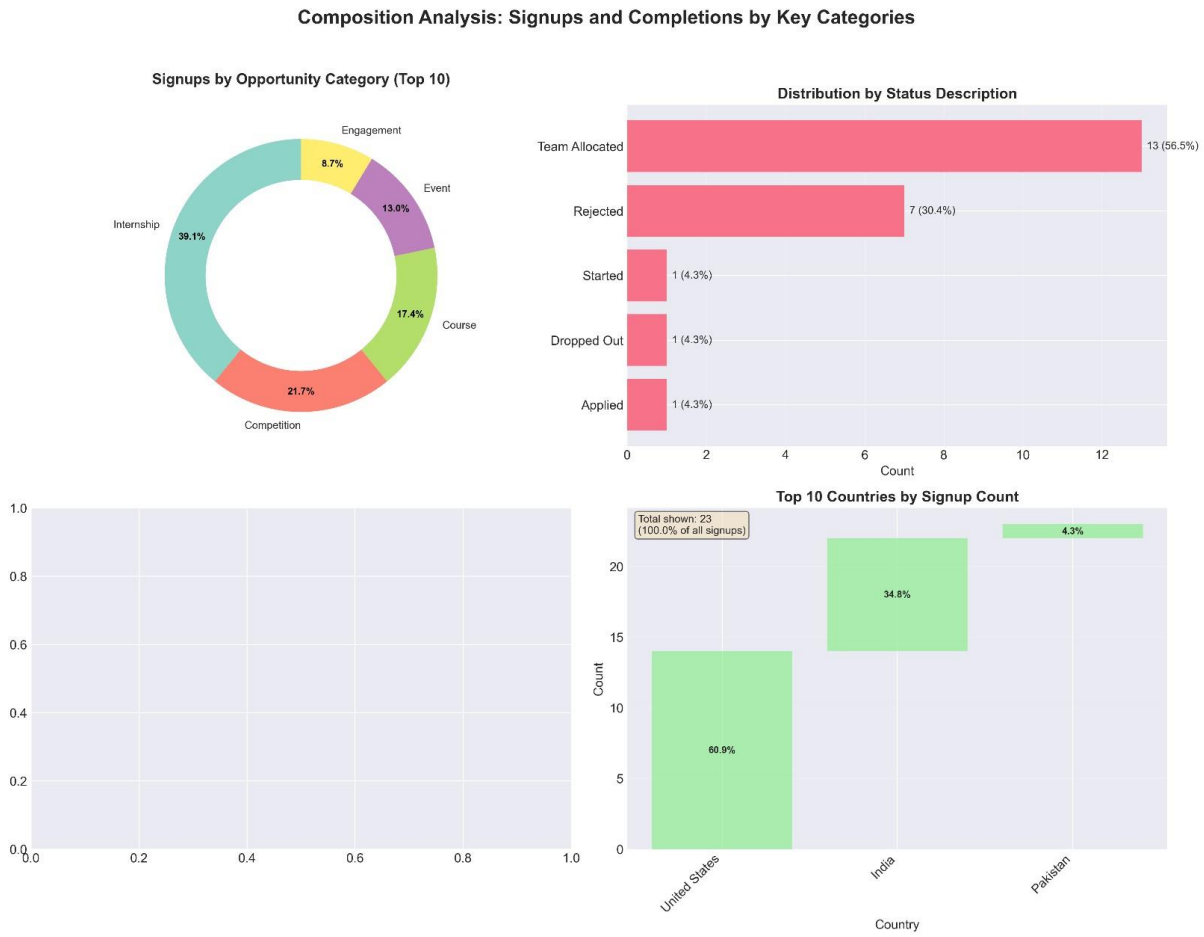
4. Key Indicators Influencing Application and Engagement Trends

The following indicators were found to be most relevant for explaining application behavior and engagement patterns:

- learner age
- speed of application after signup (days to apply)
- engagement duration and intensity
- opportunity characteristics such as category and duration

Seasonal patterns observed through signup month features indicate timing effects on application activity.

➞ Signup month trend chart
➞ Days to apply distribution
➞ Engagement segmentation chart (low / medium / high)



5. Churn Modeling and Final Model Performance (Week 3)

A churn prediction model was developed to estimate whether a learner would drop off before completing an opportunity.

Three models were evaluated:

- Logistic Regression
- Decision Tree
- Random Forest

Due to the limited dataset size, all models achieved very high performance.
The results are therefore treated as proof of concept rather than production-ready metrics.

Logistic Regression was selected as the final model because it provides strong performance and high interpretability for stakeholders.

```
==================================================
Model: Logistic Regression
==================================================

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00         3
           1       1.00      1.00      1.00         2

    accuracy                           1.00         5
   macro avg       1.00      1.00      1.00         5
weighted avg       1.00      1.00      1.00         5

ROC-AUC Score: 1.0000
Cross-validation ROC-AUC: 1.0000 (+/- 0.0000)
```

```
==================================================
Model: Decision Tree
==================================================

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00         3
           1       1.00      1.00      1.00         2

    accuracy                           1.00         5
   macro avg       1.00      1.00      1.00         5
weighted avg       1.00      1.00      1.00         5

ROC-AUC Score: 1.0000
Cross-validation ROC-AUC: 0.9333 (+/- 0.2667)
```

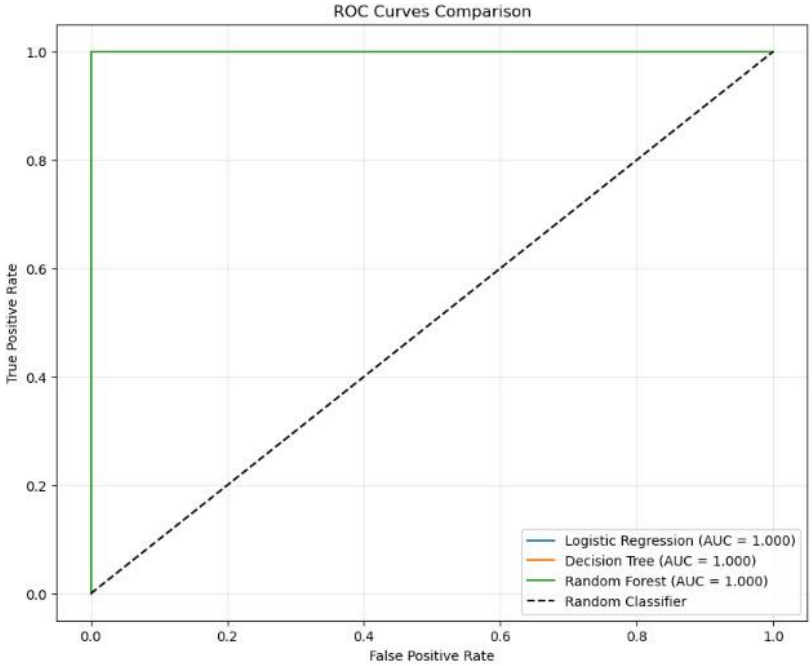
```
==================================================
Model: Random Forest
==================================================

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00         3
           1       1.00      1.00      1.00         2

    accuracy                           1.00         5
   macro avg       1.00      1.00      1.00         5
weighted avg       1.00      1.00      1.00         5

ROC-AUC Score: 1.0000
Cross-validation ROC-AUC: 1.0000 (+/- 0.0000)
```


ROC Curves Comparison
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Regression (AUC = 1.000)
Decision Tree (AUC = 1.000)
Random Forest (AUC = 1.000)
Random Classifier

6. Key Drivers of Student Churn

Model interpretation and behavioral analysis indicate that churn is most strongly associated with:

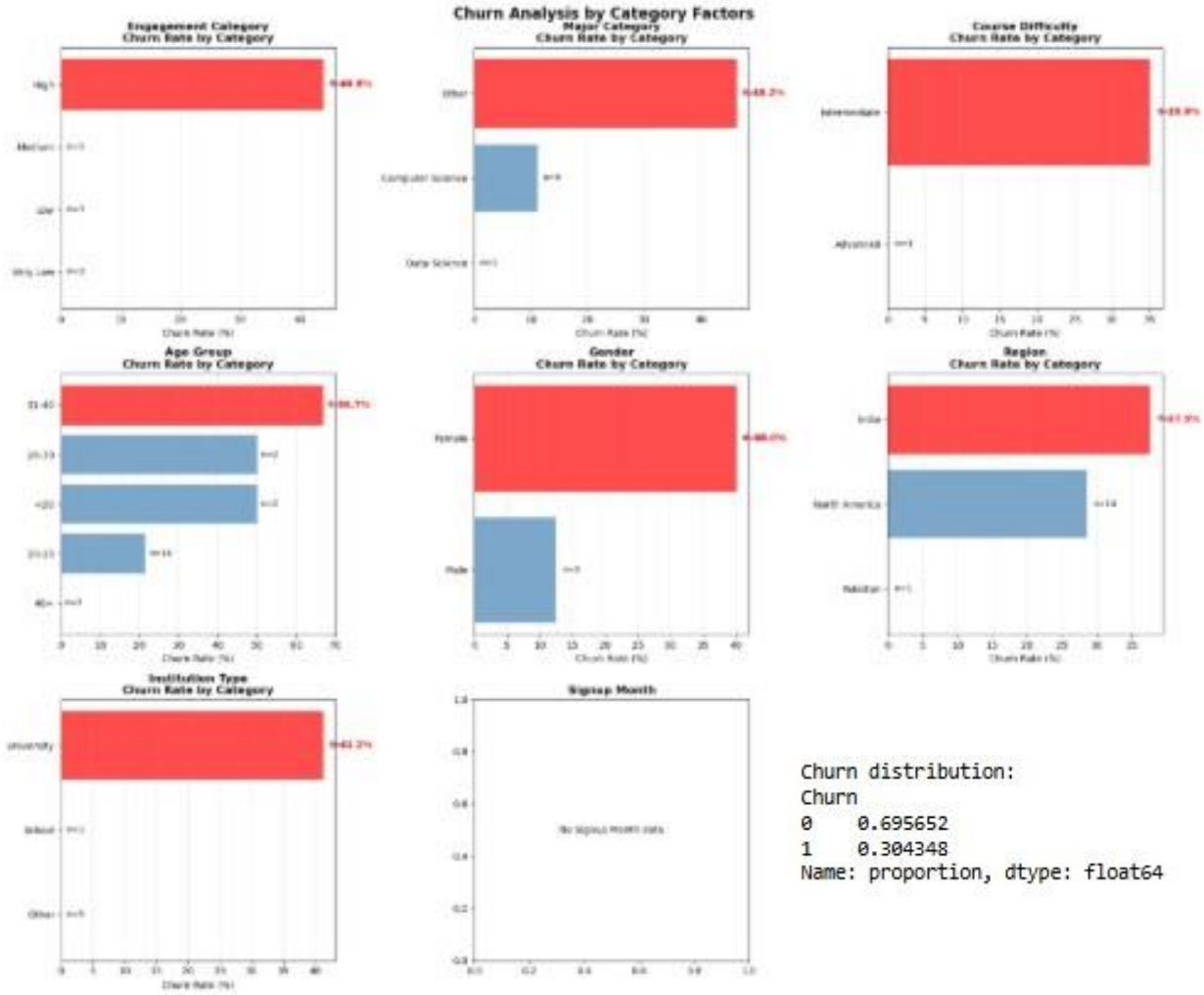
- low early engagement
- overall engagement score
- specific opportunity categories
- certain age segments

Early learner activity is the strongest indicator of future completion.

➞ Feature importance / churn driver chart
➞ Engagement vs churn plot

```
📈 CORRELATION ANALYSIS:
--------------------------------------------------
Correlation with Churn:
  Engagement_Score           0.244 (Weak Positive correlation)
  Engagement_Normalized      0.237 (Weak Positive correlation)
  Age                       -0.026 (Weak Negative correlation)
  Engagement_Days           -0.168 (Weak Negative correlation)
  Early_Drop                -0.256 (Weak Negative correlation)
```

```
VISUALIZATION OF CATEGORY ANALYSIS
```




```
Churn distribution:
Churn
0    0.695652
1    0.304348
Name: proportion, dtype: float64
```

7. Strategic Recommendations

Immediate actions

- deploy early-engagement alerts for learners with very low initial activity
- introduce guided onboarding and short learning pathways during the first days

Short-term actions

- apply category-specific interventions for programs with consistently high churn
- prioritize student support resources using churn risk scores
- design campaigns targeting underrepresented gender groups

Long-term actions

- integrate churn prediction and engagement segmentation into the platform
- redesign high-risk opportunities using behavioral feedback
- establish automated data validation rules to prevent engagement calculation errors

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8. Limitations and Risk Considerations

The analytical dataset used for modeling is small and was produced after aggressive filtering. Negative engagement values and data anomalies indicate issues in data collection or processing.

Model performance should therefore be interpreted as exploratory and educational.

Future deployments must be conducted on a validated and significantly larger dataset.

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9. Conclusion

This project demonstrates how data engineering, behavioral analytics and interpretable machine learning can be combined to support engagement and retention strategy.

With improved data pipelines and broader coverage, the proposed framework can evolve into a scalable decision-support system for student success.